

Sutapa Dey Tithi

✉ stithi@ncsu.edu | [Portfolio](#) | [Google Scholar](#) | [GitHub](#)

Research Interests

- Artificial Intelligence in Education
- Data Mining
- Collaborative AI
- Human-AI Interaction

Education

- **North Carolina State University** Raleigh, NC
Ph.D. in Computer Science, GPA: 3.97/4.0 Expected May 2027
- **North Carolina State University** Raleigh, NC
M.S. in Computer Science, GPA: 3.97/4.0 December 2025
- **Bangladesh University of Engineering and Technology** Bangladesh
B.Sc. in Computer Science and Engineering, GPA: 3.62/4.0 May 2022

Technical Skills

- **Machine Learning:** Time-series, Graph Mining, Reinforcement Learning, Machine Learning, Prompt Engineering, LLM and multi-agent LLM Evaluation, Graph RAG, Rubric-based LLM graders
- **Programming Languages Tools:** Python, R, C, C++, Java, JavaScript, MySQL, PostgreSQL, PHP, HTML, CSS, Git, Firebase, Neo4j, Jupyter, GraphViz, LaTeX
- **Libraries/Frameworks:** Pandas, NumPy, Scikit-learn, SciPy, Keras, TensorFlow, PyTorch

Research & Professional Experience

- **Graduate Research Assistant** June 2023 - Present
Advisor: Dr. Tiffany Barnes, NC State University, Raleigh, NC
 - Applied HCI principles to redesign intelligent tutoring system interface, resulting in a 50% reduction in completion time. [Poster in EDM](#)
 - Conducted user research with 2000+ undergrad students, applied Data Mining/Learning Analytics/ML and mixed-methods research techniques for designing adaptive instructional support mechanisms. [Paper in CSED](#) (Extended study under review, recommended for Round 2 in CHI'26!)
 - Developed adaptive interventions with Deep Reinforcement Learning and Bayesian knowledge tracing (ongoing analysis of collected data)
 - Advanced prompt engineering to construct logic proof solutions and hints, and built a JavaScript auto-grader to evaluate LLM-generated proofs and hints. Performed human as well as rubric-based LLM evaluation of hint explanations. Integrated a chat assistant into the existing tutor with JavaScript, Python, and Flask, and collected student-LLM conversational data in a real classroom. [Journal paper in Computers and Education: Artificial Intelligence](#)
 - Currently working towards my PhD thesis proposal, where I augment our tutor with a planning tool yielding more fine-grained interaction data to facilitate human-AI collaboration techniques utilizing DRL and/or LLM agents during problem solving.
- **Course Administrator** Summer 2024
CSC116 - Introduction to Java, NC State University, Raleigh, NC
- **Graduate Teaching Assistant** August 2022 - May 2023
CSC226 - Discrete Mathematics, NC State University, Raleigh, NC

Selected Projects

- **Automated Parkinson's Disease Assessment System:** Developed a web-based diagnostic tool for Parkinson's Disease assessment in Bangladesh, addressing healthcare accessibility challenges in low-resource settings. Collected data from 172 participants and achieved 85% accuracy with ML models for automated PD assessment. [Website](#) | [Video](#)
Technologies: React.js, Node.js, Firebase, Python, OpenCV, scikit-learn
- **Intelligent Team Formation Platform:** Built a web application enabling project-based team formation with real-time communication features. [GitHub](#) | [Demo](#)
Technologies: React, Flask, Python, PostgreSQL
- **Learn Math Educational Platform:** Created a comprehensive learning management system with rubric-based assessment, tutorial creation tools, and content quality moderation features. [GitHub](#)
Technologies: React.js, Node.js, PostgreSQL